FLUTTER-02

LAB ASSIGMENTS

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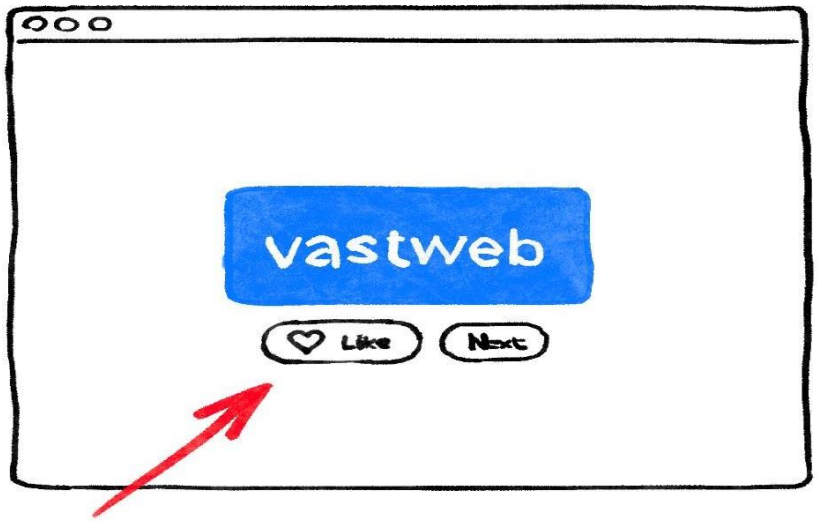
Add Functionality

The app woks, and occasionally even provides interesting word pairs. But whenever the use clicks **Next**, each word pai disappears forever. It would be better to have a way of "remembering" the best suggestions: such as a ‘Like' button.



Add the business logic

Scroll to MyAppState and add the following code:

lib/main.dart

```
// ...

class MyAppState extends ChangeNotifier {
  var current = WordPair.random();

  void getNext() {
    current = WordPair.random();
    notifyListeners();
```

```
}
// ↓ Add the code below.
var favorites = <WordPair>[];

void toggleFavorite() {
  if (favorites.contains(current)) {
    favorites.remove(current);
  } else {
    favorites.add(current);
  }
  notifyListeners();
 }
}
// ...
```

Examine the changes:

- You added a new property to `MyAppState` called `favourites`. his property is initialized with an empty list: `[]`.
- You also specified that the list can only eve contain word pairs: `<WordPair>[]`,using generics. his helps make you app more robust—Dat effuses to even *un* your app if you ty to add anything other than `WordPair` to it. In tun,you can use the `favorites` list knowing that there can never be any unwanted objects (like `null`) hiding in thee.

Note: Dart has collection types other than `List` (expressed with `[]`). You could argue that a `Set` (expressed with `{}`) would make more sense for a collection of favourites. To make this codelab as straightforward as possible, we're sticking with a list. But if you want, you can use a `Set` instead. The code wouldn't change much.

- You also added a new method, `toggleFavorite()`, which either removes the current word pair from the list of favourites (if it's already thee), adds it. In either case, the code calls `notifyListeners();` afterwards.

Add the button

With the "business logic" out of the way, it's time to work on the use interface again. Placing the 'Like' button to the left of the 'Next' button enquiries a `Row`.
he `Row` widget is the horizontal equivalent of `Column`, which you saw earlier.

Fist, wrap the existing button in a `Row`. Go to `My Homepage`'s `build ()` method, put your cursor on the `Elevated Button`, call up the **Refactor** menu with `Ctrl+ Cmd+.`, and select **Warp with Row**.

When you save, you'll notice that `Row` acts similarly to `Column`—by default, it lumps its children to the left. (`Column` lumped its children to the top.) o fix this, you could use the same approach as before, but with `mainAxisAlignment`. However, for leaning purposes, use `mainAxisSize`. his tells `Row` not to take all available horizontal space.

Make the following change:

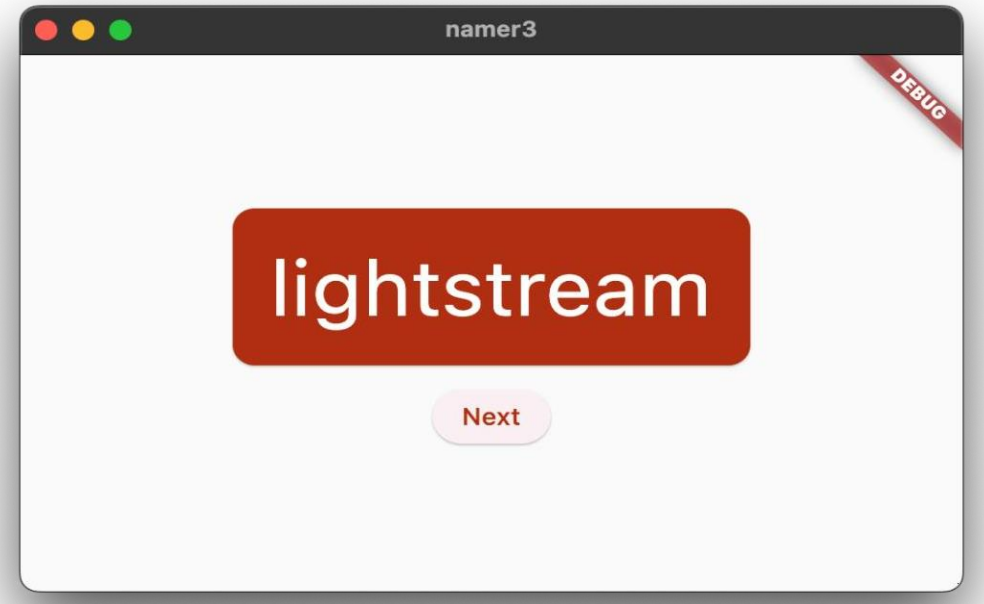
lib/main.dart

```
// ...

class MyHomePage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
```

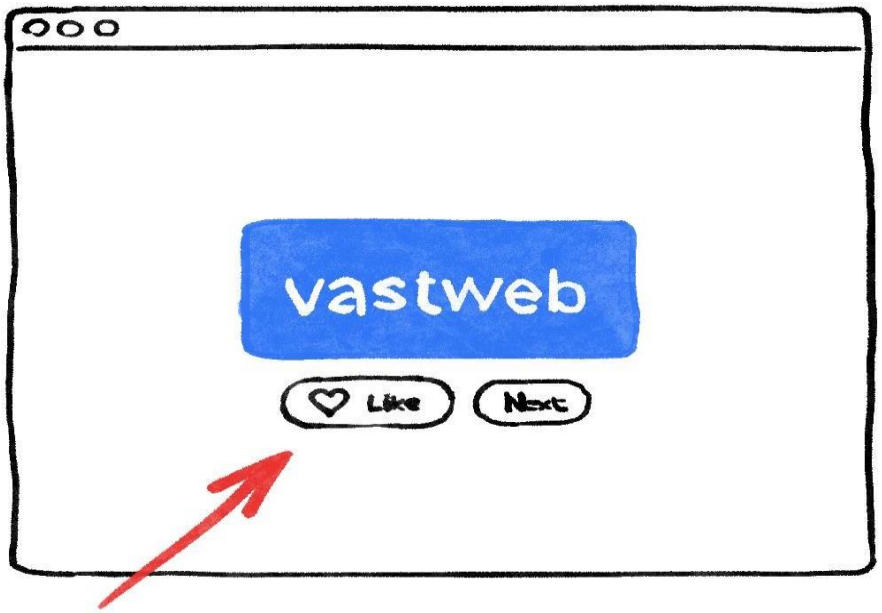
```
var appState = context.watch<MyAppState>();
var pair = appState.current;

return Scaffold(
  body: Center(
  child: Column(
     mainAxisAlignment: MainAxisAlignment.center,
     children: [
      BigCard(pair: pair),
      SizedBox(height: 10),
      Row(
       mainAxisSize: MainAxisSize.min,  // ← Add this.
       children: [
        ElevatedButton(
        onPressed: () {
          appState.getNext();
        },
        child: Text('Next'),
       ),
      ],
     ),
    ],
   ),
  ),
```

```
  );
 }
}
// ...
```

the UI is back to where it was before.



Next, add the **Like** button and connect it to `toggleFavorite()`. For a challenge, first try to do this by yourself, without looking at the code block below.



It's okay if you don't do it exactly the same way as it's done below. In fact, don't worry about the heat icon unless you early want a major challenge.

It's also completely okay to fail—this is you first hour with Flutter, after all.

Hee's one way to add the second button to `MyHomePage`. his time, use the `ElevatedButton.icon()` constucto to ceate a button with an icon. And at the top of the `build` method, choose the appopiate icon depending onwhether the current word pair is already in favourites. Also, note the use of `SizedBox` again, to keep the two buttons a bit apart.

lib/main.dart

```
// ...

class MyHomePage extends
 StatelessWidget {@override
 Widget build(BuildContext context) {
  var appState =
  context.watch<MyAppState>();var pair =
  appState.current;

  // ↓ Add this.
  IconData icon;
  if
   (appState.favorites.contains(pair))
   {icon = Icons.favorite;
  } else {
   icon = Icons.favorite_border;
  }

  return Scaffold(
   body: Center(
   child: Column(
      mainAxisAlignment:
      MainAxisAlignment.center,children: [
       BigCard(pair:
       pair),
       SizedBox(height:
       10),Row(
```

```
mainAxisSize:
MainAxisSize.min,children: [

 // ↓ And this.

 ElevatedButton.icon(
  onPressed: () {
  appState.toggleFavorite();
  },
  icon:
  Icon(icon),
  label:
  Text('Like'),
 ),
 SizedBox(width: 10),

 ElevatedButton(
  onPressed: () {
  appState.getNext();
  },
  child: Text('Next'),
 ),
 ],
 ),
 ],
 ),
 ),
 );
 }
}
```

// …